

A sample sum

A sample sum totals the observations in a sample.

$$S_n = X_1 + \cdots + X_n$$

For shipment weight, would we need an individual mass, sample mean, or sample sum?

Mean and sum

A sample sum equals the sample size times its mean.

$$S_n = n\bar{X}$$

If 40 packages have mean mass 500 g, what is their total mass?

The center of sums

For n observations from a population with mean μ , the expected sum is $n\mu$.

$$E(S_n) = n\mu$$

If one item has mean mass 50 g, what is the expected mass of 100 items?

The spread of sums

For n independent observations with common SD σ , the sum has SD $\sigma\sqrt{n}$.

$$\text{SD}(S_n) = \sigma\sqrt{n}$$

If $\sigma=4$ g and $n=100$, what is the SD of total mass?

Independent variation

Model: individual item masses have SD 4 g;
100 items are independent.

Why would using 100×4 g as the SD of total mass overstate its spread?

Increasing the total sample

Model: independent items have mean mass 50 g and SD 4 g.

How do the mean and SD of the sum change when n rises from 100 to 400?

A sum can be normal

A sum of independent normal observations is normal for every sample size.

What model describes the total of 100 independent $N(50,4)$ masses?

A normal approximation for sums

For independent, identically distributed observations with finite variance, large-sample sums can be approximated by a normal distribution.

What information about strong skewness would you want before using that approximation?

Standardizing a total

The z-score of a sum uses the sum's mean and standard deviation.

$$z = \frac{\sum x - n\mu}{\sigma\sqrt{n}}$$

For $\mu=50$, $\sigma=4$, and $n=100$, what z-score belongs to a total of 5,080 g?

A total above a limit

Model: 100 independent item masses follow $N(50,4)$, in grams.

What is the probability that their total exceeds 5,080 g?

A total below a limit

Model: 100 independent item masses follow $N(50,4)$, in grams.

What is the probability that their total is below 4,960 g?

A total within a range

Model: 100 independent item masses follow $N(50,4)$, in grams.

What is the probability that their total lies between 4,940 and 5,060 g?

A capacity cutoff

Model: 100 independent item masses follow $N(50,4)$, in grams.

What capacity would exceed the total mass in 99% of shipments?

An equivalent event

A shipment contains exactly 100 items.

What event involving the sample mean is equivalent to a total above 5,080 g?

The same standardized distance

Model: 100 independent masses follow $N(50,4)$. A total is 5,080 g.

Do the sum and its corresponding mean have different z-scores?

Sums versus averages

Model: individual masses have SD 4 g.
Compare independent samples of 25 and 100.

Does increasing n make sums or averages less variable in their original units?

A skewed population total

Model: independent task durations are exponential with mean and SD 6 minutes. A batch has 100 tasks.

What approximate normal model describes total duration?

Checking dependence

A shared machine adjustment can shift every item's mass in the same shipment.

Would the independent-sum SD formula still be justified?

Defining the correct total

Model: item masses average 50 g. Each shipment adds a fixed 200 g container.

What is the expected total mass of a container holding 100 items?